
Small Data Domain Generalization for Cross-Platform Sentiment Classification

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1 Introduction

Machine learning models are typically trained and evaluated under the assumption that training and test data come from the same distribution. In real-world applications, however, this assumption often fails. Domain generalization addresses this challenge by learning models that generalize well to unseen domains without access to target-domain data during training. A key goal is to learn invariant features, or robust and transferable representations, that capture the underlying structure of the task across domains.

In this project, we study domain generalization in data-scarce settings across multiple online platforms. We use sentiment classification as the base task and evaluate whether models trained on limited labeled data from certain domains can generalize to unseen platforms with different language styles. We evaluate across four domains that differ in formality, writing style, text length, and topics: Yelp, Amazon, Twitter (X), and Reddit. Domain shift is a major challenge in text classification because a model that performs well on one platform may fail on another due to differences in language use and platform conventions. We are also interested in how these ideas could later apply to AI-generated text detection, where distinguishing between human-written and AI-generated text has become increasingly important and increasingly difficult.

We develop a leave-one-domain-out (LODO) evaluation protocol across the four domains at four labeled data percentages: 10%, 25%, 50%, and 100%, resulting in 16 experimental conditions that examine the interaction between data scarcity and domain shift. To address cross-domain performance degradation under limited labeled data, we implement and evaluate three domain generalization techniques: text normalization, synonym augmentation, and balanced domain sampling. We also implement domain adversarial training (DANN) as an advanced domain generalization method. These methods are evaluated within a DistilBERT fine-tuning framework using cross-domain accuracy and macro F1 score. Finally, we quantify the domain generalization gap as a function of labeled data fraction, providing empirical evidence for how performance changes as training data becomes more limited.

2 Related Works

A research paper written by Dr. John Blitzer et al., titled "Biographies, Bollywood, Boom-boxes and Blenders: Domain Adaptation for Sentiment Classification," explores domain adaptation for sentiment classifiers. A relatively recent iteration of the Structural Correspondence Learning (SCL) algorithm is used to reduce relative error due to adaptation between domains by an average of 30% compared to the original SCL algorithm. Blitzer et al. also found that cross-domain accuracy does drop between 5%-10% compared to in-domain performance for product categories (books, dvds, etc.). Our work, albeit a domain generalization problem, extends the scope of the project to social media platforms, which exhibits higher levels of domain shift due to differences in text formality, length, and vocabulary. Our work also studies how the cross-domain accuracy degradation magnifies with data scarcity by training over a varied percentage of labeled data.

In a report titled "Domain Generalization with Small Data," Kecheng Chen et al. discuss domain generalization with insufficient samples, or under data scarcity, and uses a probabilistic framework to learn a domain-invariant representation of each input as a distribution rather than a fixed vector. They

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work to minimize Maximum Mean Discrepancy (MMD) between source domain distributions to reduce domain shift. This approach improved accuracy by up to 4% for less than 100 labeled training data points per domain. Our work closely aligns with their work but applied in NLP instead of in computer vision. We study how data scarcity and domain shift interact across social media domains and also focus on a small-data setting. We treat their probabilistic framework as a moonshot plan beyond what we accomplished in this report, giving way for future work.

Jason Wei and Kai Zou, in their report titled "EDA: Easy Data Augmentation Techniques for Boosting Performance on Text Classification Tasks," introduce EDA as four techniques: synonym replacement, random insertion, random swap, and random deletion. Their report shows that using EDA on smaller datasets and on 50% of labeled training data yielded higher accuracy, 88.6%, compared to an 88.3% accuracy without data augmentation and on 100% of labeled data. Using WordNet, our work directly implements the synonym replacement strategy discussed in this paper, and we evaluate its impact on cross-domain generalization under the LODO protocol, rather than single-domain evaluation.

In a report written by Jacob Devlin et al., BERT is introduced as a model designed to pretrain deep bidirectional representations from unlabeled text. We use DistilBERT in this report, a variant of BERT with slightly reduced task performance but increased inference speed and decreased memory consumption, essentially being speed and memory efficient. Our work, however, works to withhold entire domains from training and using that domain for model testing, distancing from standard BERT fine-tuning. Doing so allows us to directly measure cross-domain generalization, specifically if transformer representations learned from formal reviews can transfer to informal social media text.

Dr. Dat Quoc Nguyen, in a report titled "BERTweet: A Pre-trained Language Model for English Tweets," discusses a language model pretrained on English Twitter (X) tweets. This pretrained model performed better than general-task models, such as RoBERTa_{base}, which implies that the informal language used on Twitter are significantly different from platforms with more formal language utilized, such as Yelp and Amazon. Our work would prefer a general-purpose model such as DistilBERT without pretraining, as we focus on cross-platform generalization, in contrast to a pretrained model such as BERTweet that adapts to a single domain.

3 Method

Our main approach uses a Leave-One-Domain-Out (LODO) setup on the four sentiment datasets: Reddit, Yelp, Amazon, and Twitter. We also test our model on the Financial Phrasebank dataset from Huggingface as an out-of-domain test for our model. We sample equal numbers from each domain and maintain an equal positive/negative class balance. We train and evaluate our model under different labeled-data fractions: 10%, 25%, 50%, and 100%. For each setting, we fine-tune distilbert-base-uncased with a binary sequence classification head. Preprocessing includes lowercasing, URL and mention removal, hashtag symbol removal, repeated-character compression, emoji-to-text replacement, and synonym replacement augmentation using WordNet. The model is trained for 3 epochs using AdamW with learning rate = $1e-4$, batch size = 16, and cross-entropy loss. We evaluate both on an in-domain split, cross-domain split and on the dataset Financial PhraseBank which was not trained on, which serves as a strong out-of-domain test set because financial-news sentiment has much larger domain shift from social media and product-review text.

As our moonshot plan, we implement a Domain-Adversarial Neural Network (DANN) built on top of DistilBERT. The architecture consists of a shared DistilBERT encoder, a sentiment classification head, and a domain classification head. The sentiment head predicts binary sentiment labels (positive or negative) using labeled source-domain data, while the domain head predicts whether an input originates from the source or target domain. The core component of the DANN is the gradient reversal layer: during the forward pass, it leaves the learned representation unchanged, but during backpropagation it multiplies the gradient by $-\lambda$ (with $\lambda = 1$ in our setup). This encourages the encoder to learn domain-invariant representations that remain useful for sentiment classification while reducing domain-specific information. The DistilBERT encoder uses a hidden size of 768, corresponding to the dimensionality of the contextual sentence embeddings. Both the sentiment and domain heads include nn.Dropout(0.2) regularization layers to improve generalization. The domain classifier additionally contains an intermediate hidden layer of size 128 followed by a ReLU activation before the final domain prediction layer. Due to the additional computational cost of adversarial training, we evaluate DANN on smaller training subsets of 1000 examples each dataset

with the same fractions. The model is trained for 3 epochs to examine whether domain-adversarial learning improves cross-domain sentiment classification performance under limited-data settings.

4 Experiments

4.1 Datasets

For our experiments, we use publicly available datasets on Kaggle and Huggingface:

1. Yelp dataset
Source: yelp-dataset via kagglehub
Contains restaurant and business reviews from the Yelp Open Dataset. Reviews with 1-2 stars are labeled negative and 4-5 stars are labeled positive; 3-star reviews are excluded from the dataset.
2. Amazon dataset
Source: amazon_polarity dataset from HuggingFace
Contains amazon reviews and a label of positive or negative sentiment.
3. Reddit and Twitter dataset
Source: twitter-and-reddit-sentimental-analysis-dataset via kagglehub
Contains tweets and reddit comments made on Narendra Modi and other leaders as well as people's opinion towards the next prime minister of India. Neutral sentiments from the dataset are removed.
4. Financial Phrasebank dataset
Source: financial_phrasebank dataset via Huggingface with "75agree" configuration.
Contains sentiments from financial news with an agreement rate of 75% among annotators. Neutral sentiments are removed from the dataset.

16,000 balanced samples from each dataset are sampled, totally 64,000 samples. Each dataset is cleaned by removing NA values and mapping positive and negative labels to 1 and 0 respectively. The dataset is modified to three columns: text, which is the text of the sentiment that will be encoded by the tokenizer DistilBertTokenizerFast, a label of 0 (negative sentiment) or 1 (positive sentiment), and a domain column listing which dataset that sentiment is from.

4.2 Baselines

We compare our approach against two baseline models. The first baseline is a raw DistilBERT classifier initialized from distilbert-base-uncased with a randomly initialized binary classification head and evaluated without task-specific fine-tuning. This baseline measures how well general pretrained language representations transfer to sentiment classification without additional supervision. The second baseline is the pretrained SST-2 DistilBERT sentiment classifier (distilbert-base-uncased-finetuned-sst-2-english), which has already been fine-tuned on the Stanford Sentiment Treebank (SST-2). This provides a stronger sentiment-analysis reference point and allows us to compare our cross-domain training approach against a model trained on a large external sentiment dataset.

4.3 Metrics

We report accuracy as the primary evaluation metric and additionally macro F1 score. For each data fraction, we evaluate: (1) in-domain accuracy on held-out source-domain validation/test data, (2) cross-domain accuracy on the unseen target domain in the leave-one-domain-out setup, and (3) the domain generalization gap, defined as the difference between in-domain and cross-domain accuracy, which measures performance degradation under domain shift. During training, we also track training and validation loss and accuracy per epoch to monitor optimization behavior and potential overfitting. Together, these metrics allow us to evaluate both the robustness of our domain generalization methods and how model performance changes as available training data becomes more limited.

4.4 Base Plan Experimental Setup

Our experiment uses a LODO setup across the four datasets, holding out one domain each time as the test domain while the other three are used for training and validation with a train/test split of 90/10. We also train models with four different dataset fractions: 10%, 25%, 50%, 100%, totaling 16 experiments. Models are trained using distilbert-base-uncased with AdamW, learning rate $1e-4$, batch size 16, maximum sequence length 256, random seed 42, and 3 epochs.

4.5 Moonshot Plan Experimental Setup

For the DANN experiments, we use smaller training subsets of 1,000 examples because adversarial training is more computationally expensive than standard DistilBERT fine-tuning. For each setting, the model uses labeled source-domain data for sentiment classification and target-domain data for domain-adversarial learning. The shared DistilBERT encoder is trained with two heads: a sentiment head for positive/negative classification and a domain head for source-versus-target prediction. We use a gradient reversal layer with $\lambda = 1$, batch size = 16, AdamW optimizer, learning rate = $2e^{-5}$, and 3 epochs.

4.6 Results

4.6.1 Base Plan

Table 1: Baseline and model performance on cross-domains and the out-of-domain Financial PhraseBank dataset.

Model	Domain	Accuracy	Macro F1
Raw DistilBERT	Yelp	0.5000	0.3333
	Amazon	0.5105	0.3675
	Twitter	0.5015	0.3922
	Reddit	0.5128	0.4605
	PhraseBank	0.3221	0.2445
SST-2 DistilBERT	Yelp	0.9373	0.9372
	Amazon	0.8908	0.8906
	Twitter	0.6488	0.6308
	Reddit	0.6395	0.6126
	PhraseBank	0.7047	0.7032
Our Model	Yelp	0.5200	0.4160
	Amazon	0.5070	0.3649
	Twitter	0.5020	0.3395
	Reddit	0.5040	0.3587
	PhraseBank	0.4644	0.4586

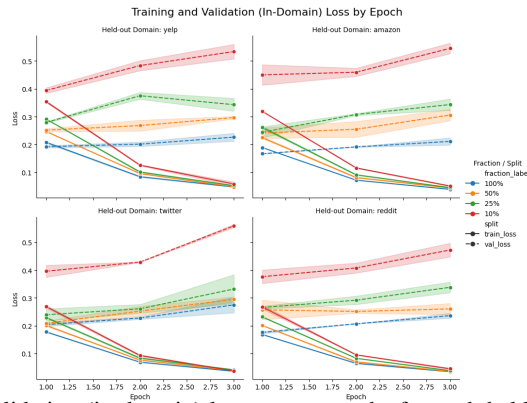


Figure 1: Training and validation (in-domain) loss across epochs for each held-out domain under different labeled-data fractions. Colors correspond to dataset fractions while line styles distinguish training and validation metrics. Shaded regions indicate confidence intervals estimated by Seaborn around the mean curves.

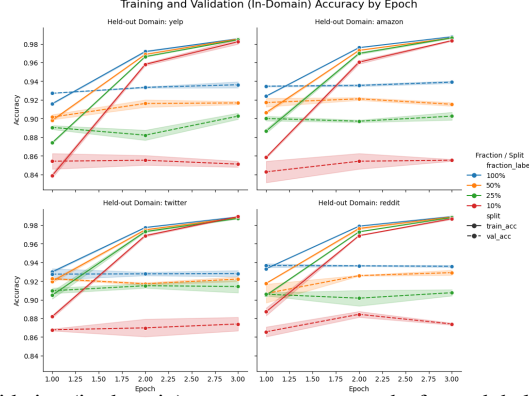


Figure 2: Training and validation (in-domain) accuracy across epochs for each held-out domain under different labeled-data fractions. Colors correspond to dataset fractions while line styles distinguish training and validation metrics. Shaded regions indicate confidence intervals estimated by Seaborn around the mean curves.

4.6.2 Moonshot Plan

Table 2: Performance of the proposed DANN-based domain adaptation model across the four domains and the Financial PhraseBank dataset.

Domain	Accuracy	Macro F1
Yelp	0.7420	0.7390
Amazon	0.7110	0.7060
Twitter	0.6520	0.6410
Reddit	0.6040	0.5960
PhraseBank	0.6280	0.6150

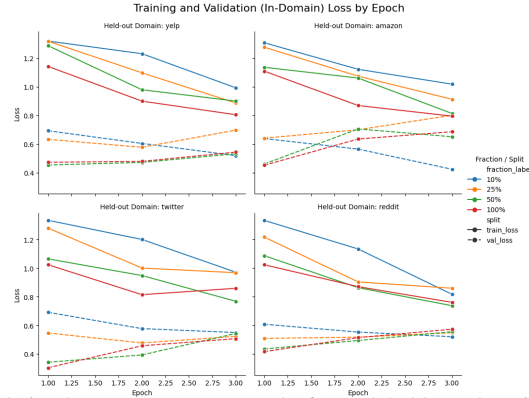


Figure 3: Training and validation loss curves across epochs for each held-out domain under different labeled-data fractions. Colors represent dataset fractions, while line styles distinguish training and validation metrics.

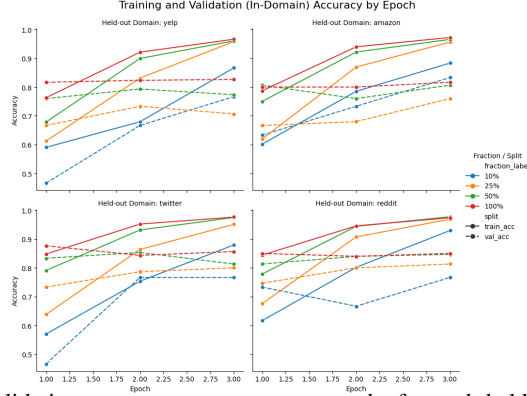


Figure 4: Training and validation accuracy curves across epochs for each held-out domain under different labeled-data fractions in the leave-one-domain-out setting. Colors represent dataset fractions, while line styles distinguish training and validation metrics.

5 Analysis

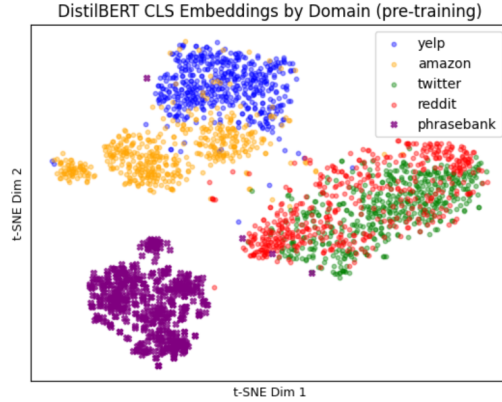


Figure 5: t-SNE visualization of DistilBERT CLS embeddings before domain-adversarial training. Samples cluster strongly by domain, indicating substantial domain-specific structure and distribution shift between Yelp, Amazon, Twitter, Reddit, and Financial PhraseBank. The clear separation between domains motivates the use of domain adaptation methods such as DANN to learn more domain-invariant representations.

The t-SNE visualization in Figure 5 shows that DistilBERT CLS embeddings cluster strongly by domain before domain-adversarial training, with Yelp, Amazon, Twitter, Reddit, and Financial PhraseBank occupying separate regions. This suggests that pretrained DistilBERT representations still contain domain-specific information, even before the sentiment classifier is trained. This helps explain why cross-domain generalization is difficult: a model may learn features that distinguish source platforms rather than features that transfer across platforms. The separation also supports our experimental results, since Yelp and Amazon are more similar review-style datasets, while Twitter and Reddit contain more slang, abbreviations, and platform-specific language. Financial PhraseBank is even more separated because financial-news sentiment differs greatly from reviews and social media text.

Table 1 shows that our model performs only slightly above the raw pretrained baseline and substantially below the SST-2 fine-tuned baseline across all domains. While our model achieves modest improvements on PhraseBank and several source domains, overall performance remains close to random guessing, suggesting limited cross-domain generalization. One possible explanation is that the low amount of training data (only 16k examples) and the low training time (3 epochs) were insufficient for the model to learn robust sentiment representations. In addition, the large domain shift between review/social-media text and financial sentiment text likely reduced transfer performance, as Financial PhraseBank contains specialized vocabulary and sentiment cues not present in the training domains. These results suggest that preprocessing and augmentation alone are not enough for domain generalization, motivating the use of more advanced approaches such as DANN.

Table 2 shows that the DANN-based approach substantially improves performance over the standard multi-domain model across all datasets. The gains are particularly noticeable on PhraseBank, where accuracy increases from near-random performance to 62.8%, suggesting that adversarial domain adaptation helps the model learn more transferable representations under larger domain shifts. Improvements on Twitter and Reddit also indicate that DANN is more effective at handling noisier and less structured text domains compared to standard fine-tuning alone.

However, performance still remains below the SST-2 baseline on several domains, particularly Yelp and Amazon. This may indicate that although DANN improves domain invariance, the relatively small training subsets and short training duration limit the model’s ability to fully learn robust sentiment representations. Nevertheless, the consistent improvements across all domains suggest that adversarial domain adaptation is a promising direction for improving cross-domain sentiment classification under limited-data settings.

Figures 1 and 2 show the in-domain training and validation behavior across different labeled-data fractions. In general, larger data fractions lead to better validation accuracy and lower validation loss, while smaller fractions show weaker and less stable performance. This suggests that increasing the amount of labeled source-domain data helps the model learn more reliable sentiment patterns.

Figures 3 and 4 show the leave-one-domain-out training and validation behavior, where the gap between training and validation performance is larger, especially for lower data fractions and harder held-out domains. In several settings, training accuracy improves while validation performance remains lower, suggesting that the model may learn source-domain surface patterns rather than invariant features.

Overall, Table 1 and Figures 1–5 show that the main challenge is not only learning sentiment, but learning invariant features that successfully transfer to unseen domains. The baseline comparison shows that sentiment-specific pretraining helps, the loss and accuracy curves show that data scarcity increases instability and overfitting, and the t-SNE visualization shows that the underlying representations already contain domain-specific structure. Together, these analyses explain why cross-domain performance is weaker for informal domains such as Twitter and Reddit and why stronger domain generalization methods may be needed.

6 Discussion

6.1 Limitations

Our approach has several limitations. First, synonym augmentation relies on WordNet, which does not capture informal language or slang, limiting its benefit for Twitter and Reddit. Second, our evaluation is restricted to binary sentiment classification; extending to finer-grained labels is a natural direction for future work. Third, we use only DistilBERT as our backbone and do not evaluate larger or domain-adapted models such as BERTweet, which may inherently handle informal text better.

6.2 Failure Modes

Our results confirm the anticipated failure modes and reveal one unexpected pattern. The largest performance degradation occurs when Twitter and Reddit, platforms with relatively high informal language usage, serve as target domains at low data fractions. At 10% labeled training data, cross-domain accuracy falls to 0.758 on Twitter and 0.747 on Reddit, compared to 0.927 and 0.887 on Yelp and Amazon respectively. While synonym augmentation is intended to increase linguistic diversity, the synonyms drawn from WordNet are formal substitutes unlikely to reflect the slang of Twitter and Reddit, limiting its effectiveness for these domains. Reddit proved slightly harder to generalize to than Twitter, which we attribute to the diversity and unpredictability of Reddit’s informal language making it harder for a model trained on formal review text to generalize to. Performance on Yelp and Amazon remained robust across all data fractions, staying above 0.88 even at 10%, reflecting the closer distributional match between these two review-style platforms.

A consistent pattern across all conditions is the large gap between training accuracy (~ 0.985) and validation/test accuracy, suggesting the model overfits to source-domain surface features rather than learning fully transferable representations. The most severe failure occurs on Financial PhraseBank, where our model achieves only 0.464 accuracy compared to 0.705 for the SST-2 DistilBERT baseline.

This indicates that our domain generalization techniques improve robustness within a family of related domains but do not confer generalization to truly out-of-distribution text such as financial news. Finally, our experiments were run on two hardware configurations (Mac MPS and HPC A100), with test accuracy differences of less than 0.01 across all conditions, confirming that results are stable and reproducible across compute environments.

6.3 Future Steps

The most immediate extension is probabilistic feature mapping, which represents inputs as distributions rather than fixed vectors and uses MMD to minimize cross-domain divergence — well-suited to our small-data setting. Additionally, our LODO protocol could be directly applied to AI-generated text detection, where the binary label represents human-written vs AI-generated text rather than sentiment, addressing the same cross-platform domain shift structure studied here.

7 Conclusion

In this project, we studied small-data domain generalization for cross-platform sentiment classification across Yelp, Amazon, Twitter, and Reddit. We used a leave-one-domain-out evaluation protocol to test whether a DistilBERT-based model trained on several source domains could generalize to an unseen target domain. We also evaluated the model under multiple labeled-data fractions, allowing us to examine how data scarcity affects cross-domain performance. Our fraction-based results show that performance was more stable on more structured review domains such as Yelp and Amazon, but dropped more noticeably on Twitter and Reddit as the amount of training data decreased. This suggests that informal social-media domains create a larger domain shift because of differences in language style, text length, slang, and platform-specific writing patterns.

Overall, our findings show that pretrained transformer models provide a strong starting point for cross-platform sentiment classification, but they do not fully solve the problem of domain shift. The decrease in performance at lower data fractions shows that data scarcity makes domain generalization more difficult, especially when the held-out domain differs significantly from the source domains. Our use of normalization, synonym replacement augmentation, balanced sampling, and DANN helped us explore different strategies for improving robustness, but the results also show that domain generalization methods depend heavily on the similarity between domains and the quality of the training data. This work highlights the importance of evaluating NLP models beyond standard in-domain accuracy because real-world models often encounter data from platforms or contexts that were not available during training. Future work could expand this project by using stronger augmentation methods, running more random seeds, testing additional domains, and exploring probabilistic or contrastive approaches for learning more robust cross-domain representations.

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